

Lab 6 – Digital to Analog Conversion

Analog & Digital Communication - EE322

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Objective

The objective of this lab is to design, compare and contrast the following two techniques of digital to analog signal:

1. Binary weighted resistors
2. R-2R ladder

Introduction

Nowadays digital circuitry is being used everywhere. Connecting the digital circuitry to sensor devices will be easy if the sensor devices are also digital. Electronic components like switches, relays, and encoders are easily interfaced with gate circuits due to the on/off nature of their signals.

In communication system, digital transmission is much faster and convenient, but the digital signals need to be converted back to analog signals at the receiving terminal. In this experiment, we will focus on the digital to analog converter, implementing two different circuit designs.

Binary Weighted Resistors DAC:

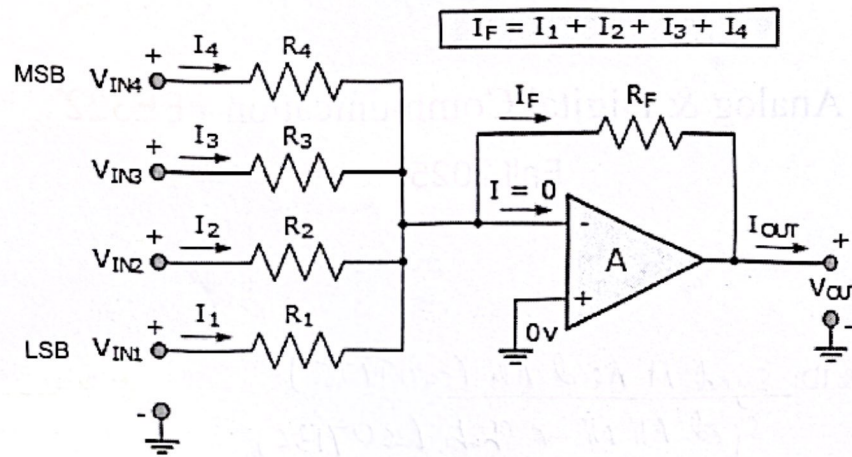


Figure 1: Binary Weighted Resistors DAC. [1]

The binary weighted resistors DAC has n number of switches, one for each bit applied to the input. The resistors of this circuit are inversely proportional to the numerical significance of its binary digit. Equation shows the computation for the output voltage of the binary weighted resistors.

$$V_{OUT} = - \left[\frac{R_F}{R_4} V_D + \frac{R_F}{R_3} V_C + \frac{R_F}{R_2} V_B + \frac{R_F}{R_1} V_A \right] \quad \dots \text{eq 1}$$

R-2R Ladder Network:

R-2R weighted resistor ladder network uses only 2 set of resistors- R and 2R. This is a 4 bit DAC. Let us consider the digital data $D_3D_2D_1D_0 = 0001$ is applied to the DAC, then the Thevenin equivalent circuit reduction is shown on next page.

V_{ref} is nothing but the input binary value voltage, that is for binary 1, $V_{ref}=5V$ and for binary 0, $V_{ref}=0V$.

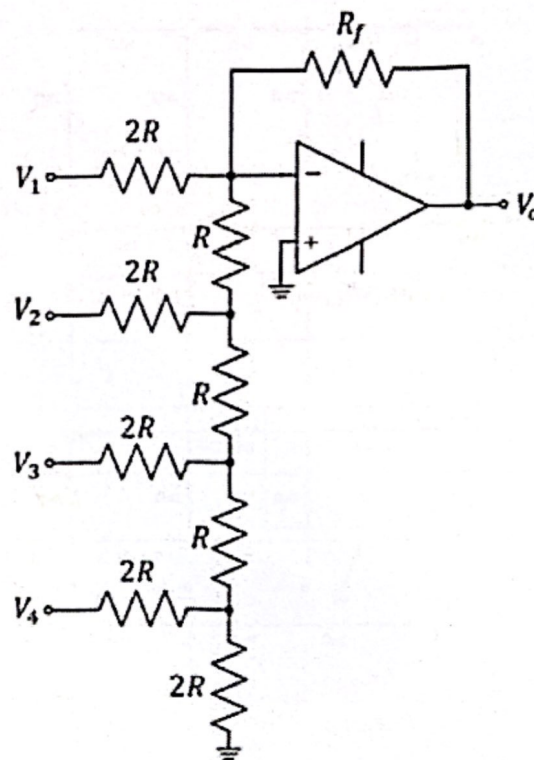


Figure 2: R-2R Ladder Network.

For 0001 only $D_0 = V_{ref}$, all other input are at 0V and can be treated as ground. So finally $V_{ref} / 16$ volt is appearing as the input to op amp. This value get multiplied by the gain of op amp circuit $-(R_f/R_i)$.

If we proceed in this manner (Thevenin equivalent reduction), we will get,

$$V_{out} = -\frac{R_f}{R_i} \left[\frac{D_0}{16} + \frac{D_1}{8} + \frac{D_2}{4} + \frac{D_3}{2} \right] V_{ref} \quad (2)$$

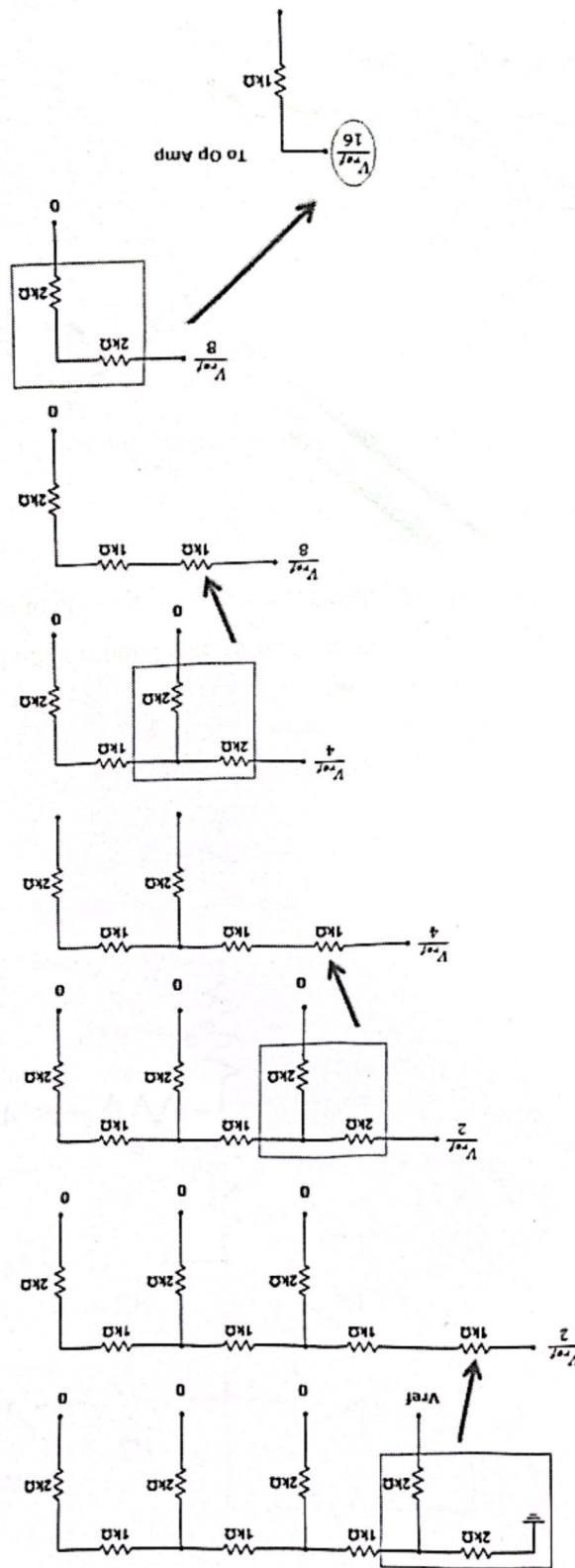


Figure 3: Thevenin equivalent circuit reduction ^[1]

Experimental Work

Apparatus Required

The following apparatus are required to perform this experiment.

No.	Apparatus	Specifications	Quantity
1	DC Power Supply	0-30 V, 1 A	1
2	Digital Multi-meter	-	1
3	Breadboard	-	1

Table 1: Apparatus required to perform this experiment.

In-Lab Tasks

Task 1: Implementing a circuit of binary weighted resistors, DAC.

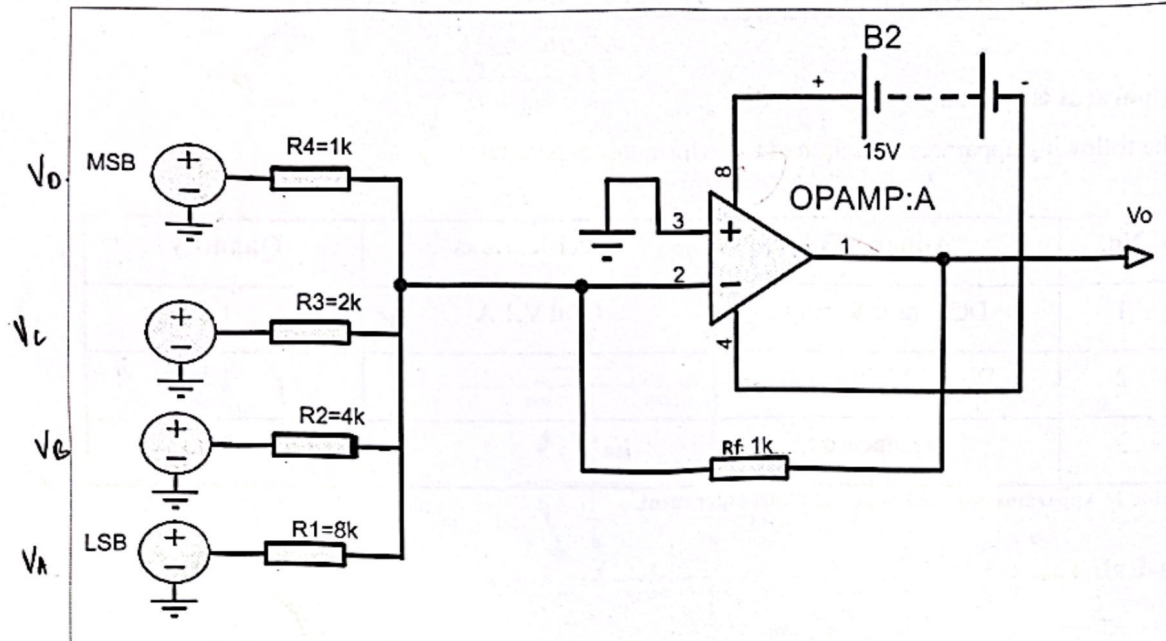
Components Required

The following components are required to perform this experiment.

No.	Components	Specifications / Values	Quantity
1	Resistors	use different resistors in combination to make binary weighted resistors	---
2	Op-Amp	LM741	1

Procedure

Implement the following circuit shown



1. Using an equation 1. calculate the theoretical values of binary weighted resistors, DAC, output and record them in.
2. Obtain all the experimental values of binary weighted resistors,
3. Measure voltage level using DMM and record them in Table
4. Now calculate percentage error in experimental value by following formula,

$$\text{Percentage Error} = \frac{\text{theoretical value} - \text{experimental value}}{\text{theoretical value}} \times 100$$

$$V_{out} = - \left[\frac{R_F}{R_4} \cdot V_D + \frac{R_F}{R_3} \cdot V_C + \frac{R_F}{R_2} \cdot V_B + \frac{R_F}{R_1} \cdot V_A \right]$$

$$\left. \begin{array}{l} * V_A = V_B = V_C = V_D = 5V \\ * R_F = 1k \end{array} \right\} V_{out} = - [V_D + 0.5V_C + 0.25V_B + 0.125V_A]$$

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* $R_4 = 1k$, $R_3 = 2k$, $R_2 = 4k$, $R_1 = 8k$.

S.No	Binary $V_A V_B V_C V_D$ 8 4 2 1	BWR theory	BWR experiment	Error (%)
0	0000	0	-0.0001	-
1	0001	-0.625	-0.6132	1.888%
2	0010	-1.25	-1.2486	0.112%
3	0011	-1.875	-1.8616	0.71467%
4	0100	-2.5 -2.5	-2.5251	-1.004%
5	0101	-3.125 -3.125	-3.1385	1.496%
6	0110	-3.75	-3.7740	-0.64%
7	0111	-4.375	-4.3852	-0.233%
8	1000	-5	-5.0582	-1.164%
9	1001	-5.625	-5.677	-0.924%
10	1010	-6.25	-6.203	0.752%
11	1011	-6.875	-6.834	0.5964%
12	1100	-7.5	-7.498	0.0267%
13	1101	-8.125	-8.107	0.2215%
14	1110	-8.75	-8.737	0.1486%
15	1111	-9.375	-9.396	-0.224%

Task 2: Implementing a circuit R-2R ladder network.

Components Required

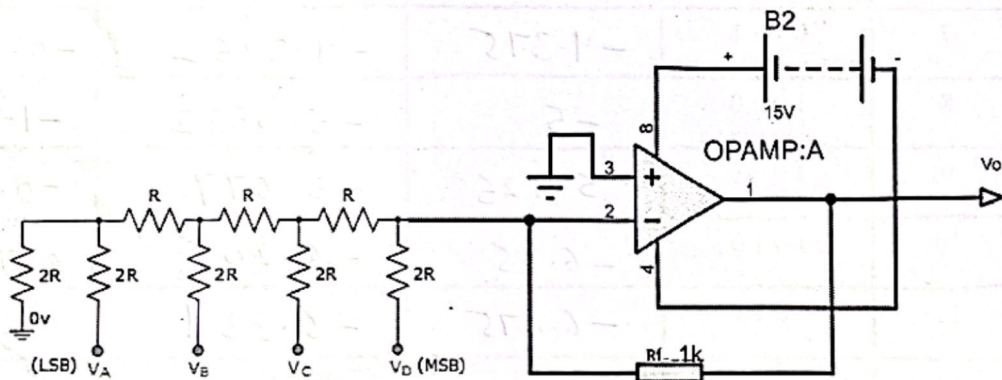
The following components are required to perform this experiment.

No.	Components	Specifications / Values	Quantity
1	Resistors	1k	Use these resistors in combination.
2	Op-Amp	LM741	1

Table 3: Components required to perform In-Lab Task 1.

Procedure

Implement the following circuit shown in Figure



1. Using an equation (2), calculate the theoretical values R2R DAC
2. Obtain all the experimental values of R-2R ladder network
3. Measure voltage level using DMM and record
4. Now calculate percentage error in experimental value by following formula,

$$\text{Percentage Error} = \frac{\text{theoretical value} - \text{experimental value}}{\text{theoretical value}} \times 100$$

Circuit : 1 2 3 4

S.No	Binary 4321	R2R theory	R2R experiment	Error (%)
0	0000	0	- 0.0001	-
1	0001	-0.3125	- 0.3148	-0.736
2	0010	-0.625	- 0.6019	3.696
3	0011	-0.9375	- 0.9131	2.603
4	0100	-1.25	- 1.2469	0.248
5	0101	-1.5625	- 1.5632	-0.0448
6	0110	-1.8750	- 1.8738	0.064
7	0111	-2.1875	- 2.1875	0
8	1000	-2.5	- 2.4905	0.38
9	1001	-2.8125	- 2.8042	0.2951
10	1010	-3.125	- 3.1133	0.3744
11	1011	-3.4375	- 3.4250	0.3636
12	1100	-3.75	- 3.7371	0.344
13	1101	-4.0625	- 4.0496	0.3175
14	1110	-4.375	- 4.3588	0.3703
15	1111	-4.6875	- 4.6692	0.3904

Conclusion:

Compare both digital to analog converters in terms of design and performance.

What is the voltage step size in task 1 and task 2
